

Question and design

Can prolonged SFT teach a model to combine multiple structural hypotheses, then improve its histories through rejection selection?

Warm up the multi format and explicit final-prediction token. Later use hypothesis weight 0.1 with full reference-answer supervision and refreshed corpora. Rejection ranks generated final answers but trains against reference contacts.

First full-model trial completed

Current exp232-derived 1.5B model; one 8-H100 node on cw-us-east-02a, with all large I/O in its existing S3 bucket. Iris selected placement from live fleet capacity.

2,048 source-balanced proteins: 2,023 training, 25 sequence-hash-held-out. Sixteen bootstrap drafts, 50% plain rehearsal, hypothesis weight 1.0. Global batch 32, 256 optimizer steps, 8,192 training documents processed.

W&B: open-athena/MarinFold/exp281-trial-s03. Source: bab3f50a. Final model: trial-s03/checkpoints/exp281-trial-s03/step-256 under the exp281 S3 prefix.

Result: format gate failed

At step 256, sample eight completions per held-out protein in each mode. Natural seed 281; forced seed 282 and budgets 0/256/1024/2048.

Natural: 0/200 valid. Forced: 14/200 valid (7%). Both fall below the fixed 99% validity gate. The natural multiple-nonempty-hypothesis gate also fails.

188/200 natural samples contain multiple raw section markers, but only 8/200 contain the final marker. In forced mode, 185/200 restart a hypothesis after the final marker. Repeated sections have appeared; reliable finalization has not.

Lower loss did not imply finalization

Internal teacher-forced validation loss decreases from 2.781 at step 32 to 2.244 at step 256. Final training minibatch loss: 1.869.

The corpus has only 500 supervised natural final markers per pass; the other 1,054 supervised markers are plain-mode starts. Aggregate contact-token loss can improve while the rare switching behavior remains unreliable.

Invalid trajectories retain zero final F1: natural 0.0000, forced 0.03193. These are engineering/format diagnostics, not evidence of improved protein accuracy.

Throughput and recovery

Peak allocated memory: 35.45 GB per H100. Median optimizer step: 1.049 seconds, excluding checkpoint publication and validation. A separate eight-step batch-8 profile reached 58,480 aggregate tokens/s after step 1.

The original trial stalled publishing its step-128 optimizer, then a waiting rank aborted. Resume from complete step 64 succeeded through step 256. Long runs need bounded checkpoint retries. The canonical training CSV uses the successful resumed trajectory; W&B explicit-step logging suppressed replayed metrics after rollback.

Next gate and open research questions

Extend format warm-up toward the proposed 2,000 steps and track final-marker/termination losses separately before synthesis or rejection training. A short-history curriculum and stronger transition supervision are additional candidate interventions. This 256-step pilot does not test the intended long SFT horizon.

The first trial establishes full-model execution and checkpoint recovery. It does not establish useful diversity, improved contact accuracy, or the benefit of refreshed SFT or rejection selection. FoldBench eval-test remains untouched.

The public HF report and committed per-input metrics, diagnostics, timing CSVs, and plotting inputs preserve this negative result for reproduction.

Continuation: overfitting through step 1,750

Continue step 256 to 2,000 on the same corpus and eight-H100 placement. Preserve Adam state, data positions and RNG; rewarm LR over 100 steps, then decay. This isolates additional format training from changes to hypothesis count, sampling or loss weights.

New diagnostics separate natural final-marker, final-answer and termination losses. Publication runs in a killable process with a five-minute deadline; complete checkpoints cannot be overwritten. Nineteen tests and real CPU/GPU continuation-resume checks pass.

Training loss reaches 0.0024 while held-out loss rises from 2.2445 to 5.9468. Natural final-marker and final-answer termination accuracy remain zero in the sparse teacher-forced checks (six markers and nine end tokens). This is pronounced overfitting of the frozen pilot corpus, not a test of larger refreshed SFT or rejection training.

Continuation recovery is queued

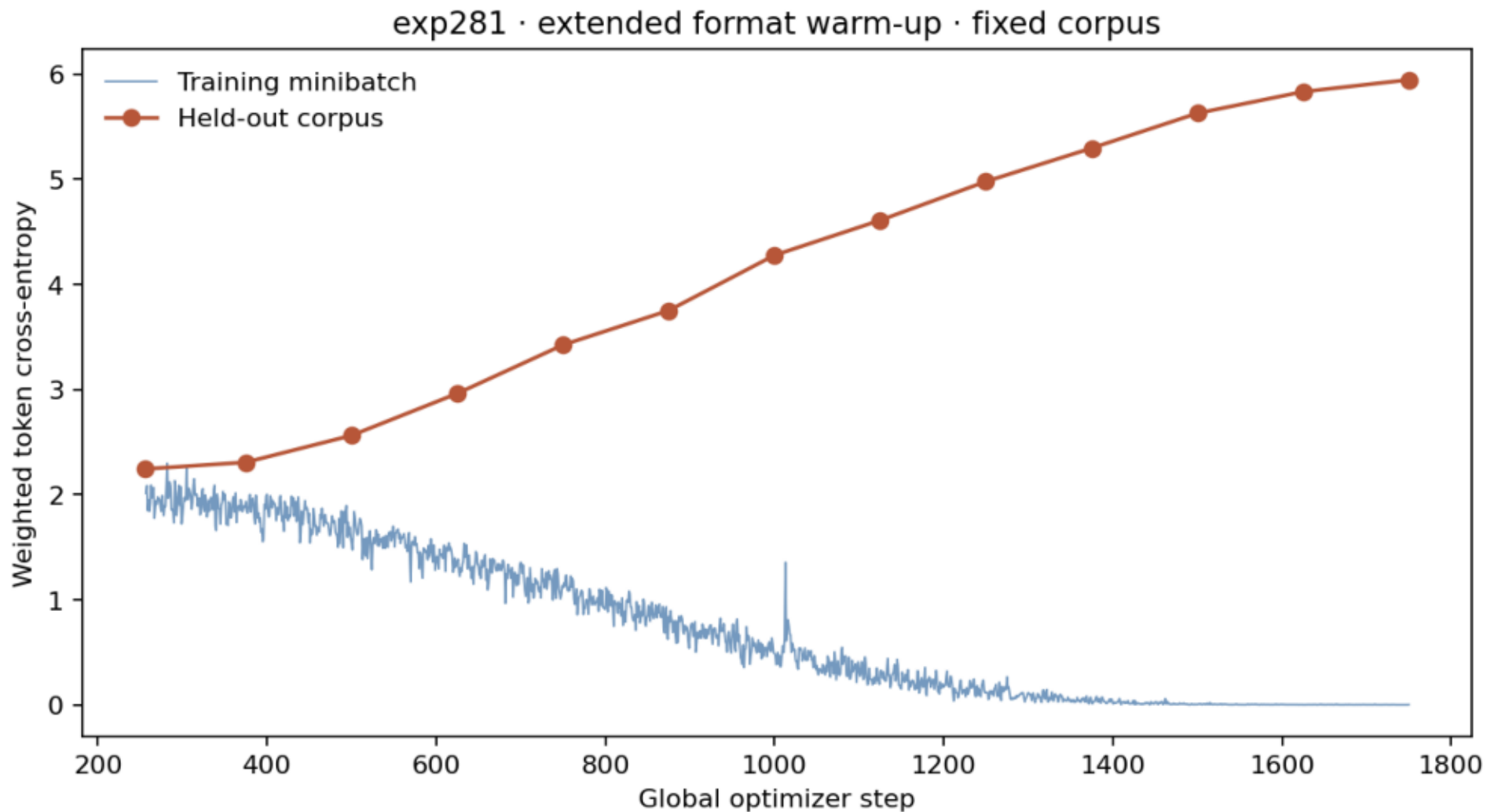
Step-1,750 checkpoint publication exceeded its 300-second deadline and failed the first attempt promptly. Five earlier checkpoints saved in 97-101 seconds; step 1,500 is complete. Cleanup of the failed attempt's abandoned uploads succeeded.

Exact resume /bizon/exp281-format-s02-r1 is queued in US-EAST-02A as of September 9, 23:43 UTC. RNO2A has capacity, but moving the recovery requires approval for approximately 53 GB of cross-region checkpoint I/O.

Step 2,000 and its evaluation remain pending. Final evaluation will use the same 25 proteins and 400 total completions as the pilot. Source: 4721c76c. No synthesis or rejection training has started.

format_s02_loss.png

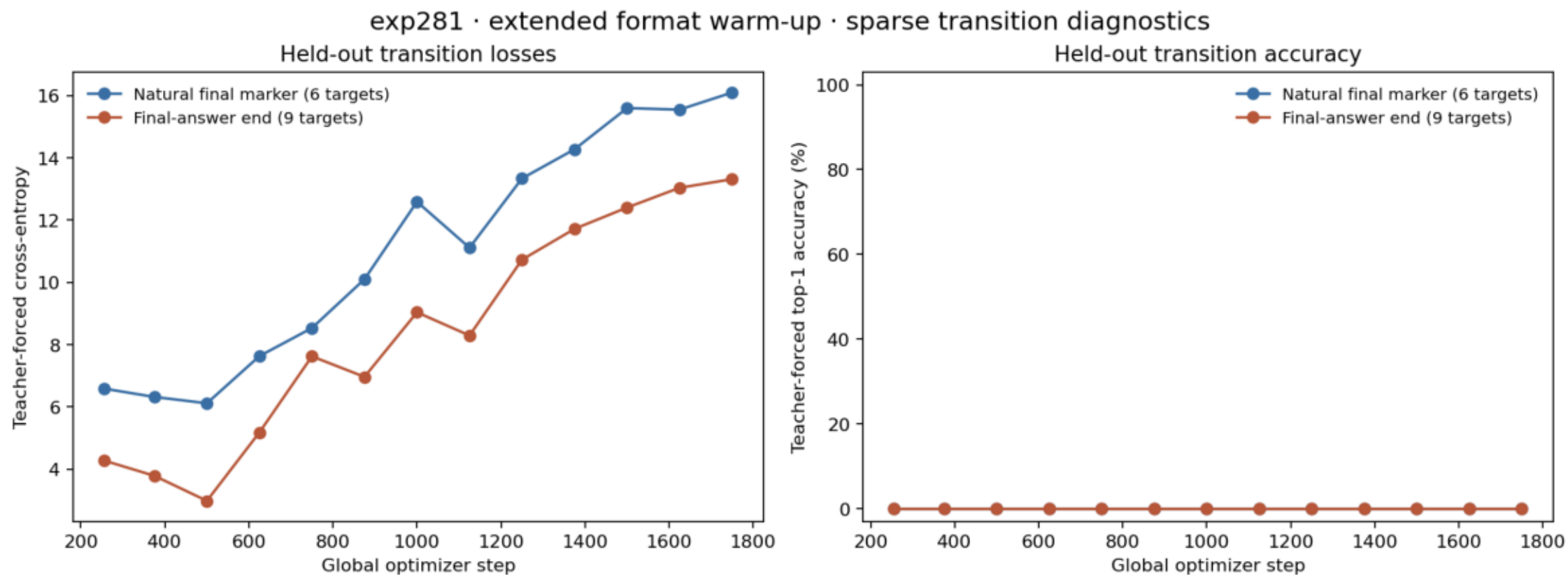
Same frozen 2,023-document training and 25-document held-out corpora. Training loss approaches zero while held-out loss rises. These token losses do not measure free-running contact accuracy.



```
$ uv run --no-project --with matplotlib python -m experiments.exp281_models_iterated_sft_and_rejection_fine_tuning._scripts.plot_continuation
```

format_s02_transitions.png

The same held-out corpus supplies six supervised natural final markers and nine multi-document end tokens at every check. Forced markers and padding are excluded. These sparse teacher-forced diagnostics do not replace the separate free-running format gate.



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$ uv run --no-project --with matplotlib python -m experiments.exp281_models_iterated_sft_and_rejection_fine_tuning._scripts.plot_continuation
```

trial_s03.png

Loss improves but the fixed format gate fails on 25 held-out proteins, eight samples each. Invalid outputs remain in the denominator. Training CSV uses the successful resumed trajectory; the lost attempt's replayed steps are excluded. No protein-accuracy improvement is established.

